Marker–Aligned Subspace Separation for Implicit Sentiment Analysis

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Abstract. Implicit Sentiment Analysis (ISA) is a challenging subtask of Aspect-Based Sentiment Analysis (ABSA), as sentiment toward an aspect term is conveyed through contextual, pragmatic, or domain-specific cues rather than explicit opinion expressions that current approaches still struggle to capture. In this study, we propose Marker-Aligned Subspace Separation (MASS), a fine-tuning framework that combines aspect marking with subjectivity–polarity subspace decomposition. MASS first derives an aspect-focused contextual representation from the marked input, then projects it into separate subjectivity and polarity subspaces for hierarchical Neutral/Polar and Positive/Negative prediction, reducing interference between the two objectives. Experiments using the Flan-T5 backbone on the Laptop and Restaurant benchmarks show that MASS achieves higher performance than competitive baselines, with an absolute accuracy gain of 4.46 points on the implicit-only Restaurant test set, while ablation study results demonstrate that each component contributes to the overall performance improvement. Our source code is available at the GitHub repository[1].

Keywords: Sentiment Analysis · Large Language Models · Fine-tuning.

1 Introduction

Unlike traditional Sentiment Analysis (SA) relying on explicit lexical cues, ISA extracts sentiment through nuanced reasoning over semantics, pragmatics, and cultural context [18]. While crucial for interpreting widespread indirect expressions on social media, ISA remains challenging due to the lack of explicit sentiment words.

Despite Large Language Models (LLMs) advancing in general SA [26,24,7], progress in ISA remains constrained by the limitations of current approaches. Prompting-based methods [6,16] are adaptable but often yield suboptimal results due to error-prone reasoning chains that miss ISA nuances. Conversely, fine-tuning approaches using models like Flan-T5 [21,8,9] perform better but largely depend on LLM-driven data augmentation, introducing prohibitive costs and synthetic bias risks. This necessitates a new fine-tuning strategy that maximizes existing data utility without relying on costly auxiliary generation.

Our analysis in Section 2 identifies two main challenges in ISA. First, limited focus on the aspect term may hinder the association between contextual sentiment cues

[1] https://github.com/ngmhoang62/MASS-ISA

and the target aspect. Second, the weak distinction between Neutral and Polar instances leads to frequent errors at the Neutral–Polar boundary. Existing studies offer several relevant insights into these issues. Aspect markers can help direct model attention toward target aspect terms in noisy contexts [1], while hierarchical SA has shown the benefit of distinguishing subjective from objective content before polarity classification [5,14]. Orthogonal constraints on hidden representations or parameter subspaces can further encourage different components to capture complementary information and reduce task interference [25]. Inspired by these findings, we make the following contributions:

1. We introduce MASS, which combines marker-based aspect representations with hierarchical Neutral–Polar and Positive–Negative prediction while jointly learning separate subjectivity and polarity subspaces, yielding consistent improvements on the Laptop and Restaurant datasets [10].
2. We analyze the results, conduct statistical tests, and perform an ablation study to better understand the improvements of the proposed method.

2 Analysis of Limitations in Prior Work

Table 1. Confusion matrices for THOR with Flan-T5-Base on the full test sets. Rows are true labels (T), columns are predicted labels (P).

THOR – Restaurant			
T/P	**Neg**	**Neu**	**Pos**
Neg	**168**	13	15
Neu	24	**87**	85
Pos	10	8	**710**

THOR – Laptop			
T/P	**Neg**	**Neu**	**Pos**
Neg	**110**	9	9
Neu	40	**94**	35
Pos	11	19	**311**

To examine the limitations of prior approaches, we analyze THOR [6], the most recent framework with publicly available code, from two perspectives: attention to the aspect term and error distributions in the confusion matrices. On the full test sets, THOR exhibits low aspect attention ratio (AAR). AAR is computed by averaging attention across all layers and heads and measuring the proportion of the resulting attention weights assigned to the aspect term tokens, normalized over non-padding input tokens. THOR obtains an AAR of 4.99% on Laptop and 6.54% on Restaurant. Although attention weights provide only an approximate indication of token contribution, these results suggest that improving aspect focus remains a meaningful direction. The confusion matrices in Table 1 show that errors involving Neutral—i.e., off-diagonal entries in its row or column—account for 83.74–83.87% of all errors across the two domains, or 83.81% when pooled (233/278). This pattern indicates that most errors occur at the Neutral–Polar boundary. **Conclusion:** THOR shows limited aspect focus and frequent confusion between Neutral and Polar instances, motivating the use of aspect markers and Subjectivity-Polarity decomposition in our MASS.

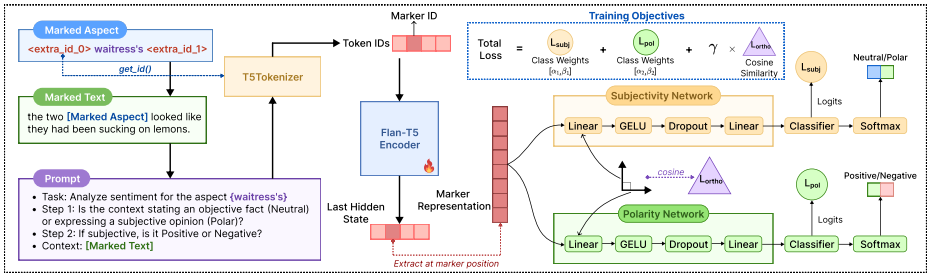


Fig. 1. The overall architecture of our proposed MASS framework.

3 Related Works

Sentiment Analysis. LLMs now represent the prevailing paradigm in SA. A comprehensive study by [26] highlights LLMs' versatility across various tasks, ranging from standard classification to intricate aspect-based and multifaceted evaluations. To address data sparsity in low-resource scenarios, cross-lingual SA has emerged as a prominent direction, utilizing knowledge transfer to improve model generalization [20]. Additionally, recent prompting-based frameworks have integrated in-context learning and reasoning strategies to better align LLM outputs with the specific demands of sentiment inference [24,19]. Furthermore, [23] introduces the CompEffDist framework to develop compact yet powerful SA models through automated instruction generation and intelligent data filtering, enabling a 3B parameters model to match the performance of a teacher model 20 times its size using only 10% of the training data.

Implicit Sentiment Analysis. ISA presents greater challenges than conventional SA by requiring the inference of unexpressed sentiment. Foundational benchmarks include aspect-level datasets for English laptop and restaurant domains [10] and a sentence-level Vietnamese dataset for technology reviews [11]. Subsequent research has bifurcated into two main directions: supervised fine-tuning approaches, primarily based on Flan-T5 backbone [21,8,9], and prompting-based methods utilizing multi-hop reasoning strategies to leverage LLM's reasoning [6,16]. While prompting methods are bounded by their purely in-context nature without domain pre-training, fine-tuning approaches—despite higher accuracy—are heavily bottle-necked by labeled data scarcity and the need for LLM-based data augmentation, constraining their domain adaptability.

4 Methodology

4.1 Task Definition and Method Overview

In this work, we focus on aspect-based sentiment classification, a fundamental subtask of ABSA. Given a dataset of N instances, $\mathcal{D} = (x_i, a_i, y_i)_{i=1}^{N}$, where x_i is an input sentence, a_i is the target aspect term occurring in x_i, and $y_i \in \mathcal{Y}$ is its corresponding ground-truth label, the task aims to predict the sentiment polarity expressed toward a_i:

$$f : (x_i, a_i) \mapsto \hat{y}_i, \qquad \hat{y}_i \in \mathcal{Y}, \tag{1}$$

where $\mathcal{Y} = \{\text{Positive}, \text{Negative}, \text{Neutral}\}$ denotes the sentiment label set.

We propose a hierarchical framework designed for the Flan-T5 backbone, which has been widely adopted in prior studies and thus supports fair comparison. It decomposes three-class sentiment prediction into subjectivity and polarity classification, using special marker tokens to identify the aspect term and separate projections to derive the corresponding representations, as illustrated in Figure 1.

4.2 Aspect Marker and Prompt Construction

Initially, we formulate a structured prompt with two instruction steps by combining the *Marked Text* and the *Marked Aspect*. The aspect term is wrapped with special sentinel tokens (`<extra_id_0>` and `<extra_id_1>`) to indicate its boundaries and then inserted back into the original sentence to construct the *Marked Text*. The two steps first distinguish Neutral from Polar sentiment and then classify Polar instances as Positive or Negative. This design leverages the contextual encoding capability of Flan-T5 to derive aspect-specific semantic representations, which subsequently support the separation of subjectivity and polarity subspaces.

4.3 Marker Representation Extraction

The prompt is tokenized using the T5Tokenizer and processed solely by the Flan-T5 encoder, without employing the decoder. The encoder is fine-tuned in a standard manner on the downstream ABSA datasets to capture domain-specific sentiment patterns. Rather than applying sequence-wide pooling, we track the position m of the special `<extra_id_0>` sentinel token, referred to as the marker position, to obtain a representation centered on the aspect term.

Formally, let $\mathcal{H} \in \mathbb{R}^{L \times d}$ denote the *Last Hidden State* matrix of the encoder, where L is the sequence length and d is the hidden dimension. The vector at position m in $\mathcal{H}$ is extracted as the aspect-specific *Marker Representation*, denoted by $\mathbf{h}_{\text{marker}} \in \mathbb{R}^d$. This representation summarizes the prompt context with emphasis on the marked aspect and is subsequently passed to the subjectivity and polarity networks for sentiment modeling.

4.4 Subjectivity-Polarity Subspace Decomposition

The marker representation $\mathbf{h}_{\text{marker}}$ is projected into two separate representation spaces through the *Subjectivity* and *Polarity* networks. This decomposition reflects the distinction between cues indicating whether sentiment is expressed and those determining its positive or negative orientation. During training, both branches are optimized jointly in an end-to-end manner, allowing the subjectivity, polarity, and orthogonality losses to update the shared encoder and their respective projection networks.

During inference, prediction follows a hierarchical procedure. Let $\mathcal{S} = \{\text{Neutral}, \text{Polar}\}$ and $\mathcal{P} = \{\text{Positive}, \text{Negative}\}$ denote the subjectivity and polarity label spaces, respectively. The subjectivity prediction is first obtained as:

$$\hat{s}_i = \arg\max_{s \in \mathcal{S}} p_{\text{subj}}(s \mid \mathbf{h}_{\text{marker},i}). \tag{2}$$

The final sentiment prediction is then determined by:

$$\hat{y}_i = \begin{cases} \text{Neutral}, & \text{if } \hat{s}_i = \text{Neutral}, \\ \arg\max_{p \in \mathcal{P}} p_{\text{pol}}(p \mid \mathbf{h}_{\text{marker},i}), & \text{if } \hat{s}_i = \text{Polar}. \end{cases} \tag{3}$$

Thus, the polarity network is evaluated only for instances predicted as Polar.

As an auxiliary regularizer, we apply a cosine-based orthogonality constraint to the first linear projection layers of the two networks. Let $\mathbf{W}_{\text{subj}}$ and $\mathbf{W}_{\text{pol}} \in \mathbb{R}^{d' \times d}$ denote the corresponding projection matrices, where d and d' are the encoder hidden dimension and reduced projection dimension, respectively. After row-wise L_2 normalization, yielding $\tilde{\mathbf{W}}_{\text{subj}}$ and $\tilde{\mathbf{W}}_{\text{pol}}$, the orthogonality penalty is defined as:

$$\mathcal{L}_{\text{ortho}} = \left\| \tilde{\mathbf{W}}_{\text{subj}} \tilde{\mathbf{W}}_{\text{pol}}^{\top} \right\|_F. \tag{4}$$

This penalty encourages the two projection spaces to capture less overlapping information. Following the projection stage, both branches apply identical *GELU* $\rightarrow$ *Dropout* $\rightarrow$ *Linear* $\rightarrow$ *Classifier* pipelines to produce logits for their respective classification objectives.

4.5 Training Objectives

The framework is optimized using the following joint training objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{subj}} + \mathcal{L}_{\text{pol}} + \gamma \mathcal{L}_{\text{ortho}}, \tag{5}$$

where γ controls the contribution of the orthogonality regularizer.

To account for class imbalance, the subjectivity and polarity objectives are formulated as weighted cross-entropy losses:

$$\mathcal{L}_{\text{subj}} = -\frac{1}{N} \sum_{i=1}^{N} \left(\alpha_1 y_i^{\text{neu}} \log q_i^{\text{neu}} + \beta_1 y_i^{\text{pol}} \log q_i^{\text{pol}} \right) \tag{6}$$

$$\mathcal{L}_{\text{pol}} = -\frac{1}{N_{\text{polar}}} \sum_{j=1}^{N_{\text{polar}}} \left(\alpha_2 y_j^{\text{pos}} \log q_j^{\text{pos}} + \beta_2 y_j^{\text{neg}} \log q_j^{\text{neg}} \right) \tag{7}$$

where y denotes the one-hot ground truth labels and q denotes the predicted class probabilities. The polarity loss $\mathcal{L}_{\text{pol}}$ is computed only over the N_{polar} non-neutral training instances. The class weights are determined from inverse class frequencies, such that $\frac{\alpha_1}{\beta_1} = \frac{N_{\text{polar}}}{N_{\text{neutral}}}$ for subjectivity classification and $\frac{\alpha_2}{\beta_2} = \frac{N_{\text{negative}}}{N_{\text{positive}}}$ for polarity classification. Here, N_{neutral}, N_{positive}, and N_{negative} denote the corresponding class counts in the training set.

5 Experiments

5.1 Datasets

We conduct experiments on the Laptop and Restaurant ABSA datasets originally introduced in SemEval-2014 Task 4 [15]. [10] later extended these datasets by assigning

Table 2. Statistics of the Laptop and Restaurant datasets.

Split	Total	Positive	Negative	Neutral	Implicit (%)
			Laptop		
Train	2313	987	866	460	716 (30.96%)
Test	638	341	128	169	175 (27.43%)
Total	2951	1328	994	629	891 (30.19%)
			Restaurant		
Train	3602	2164	805	633	1030 (28.60%)
Test	1120	728	196	196	267 (23.84%)
Total	4722	2892	1001	829	1297 (27.47%)

each aspect term a boolean implicit label. This annotation indicates whether the sentiment toward each aspect term is implicit, and allows implicit instances to be isolated for separate ISA evaluation. We use these extended datasets because they are established benchmarks in both ABSA and ISA tasks, and enable direct comparison with prior studies.

Table 2 reports the train and test sizes, sentiment-label distributions, and proportions of implicit instances for both domains. During inference, we evaluate on both the complete test set, denoted as *All*, and a domain-specific subset containing only implicit sentiment instances. Hereafter, these subsets are referred to as the ISA subsets, or simply *ISA*, following the evaluation settings adopted in prior works. In addition, the original datasets also contain a *conflict* label, defined as a case in which "both positive and negative sentiment is expressed about an aspect term" [15]. Following previous publicly available implementations [10,6], we remove conflict-labeled instances from both the training and test sets to ensure consistent and fair comparison.

5.2 Metrics

We evaluate our proposed approach using Accuracy and Macro-F1 (denoted as *Acc.* and *MF1*, respectively). Together, these metrics measure overall performance while accounting for all classes under the substantial label imbalance in the datasets, and enable direct comparison with prior works. Additionally, we report Macro-F1 on the ISA subsets, a metric often omitted in prior studies, to specifically assess performance on ISA.

5.3 Models

For our experiments, we employ the Flan-T5 architecture as the backbone, specifically utilizing two variants: Flan-T5-Base[2] (250M) and Flan-T5-XXL[3] (11B). These models are widely used as ISA backbones, enabling direct and fair comparison with prior studies. To provide a broader comparative perspective, we conduct prompt-based supervised

[2] https://huggingface.co/google/flan-t5-base
[3] https://huggingface.co/google/flan-t5-xxl

fine-tuning on 8B-parameter LLMs, including Llama-3.1-8B-Instruct[4] and Qwen3-8B[5]. We also include closed-source models GPT-4o and GPT-4.1 as training-free prompting baselines.

5.4 Baselines

Our evaluated baselines are categorized into two categories:

Training-Free Prompting Baselines. We evaluate zero-shot, few-shot, and Chain-of-Thought (CoT) prompting with GPT-4.1. CoT extends zero-shot prompting with the trigger *"Let's think step by step."*, while few-shot prompting uses three examples sampled from the corresponding training set.

Supervised Fine-Tuning Approaches.

BERT and T5. A widely used set of supervised fine-tuning baselines based on BERT-Base (110M) and T5-Base (220M): SPC [4], ADA [17], RGAT [22], CEPT/SCAPT [10], and ABSA-ESA [13]. As these methods are well-established and adopted in prior works, we use their published results compiled by MT-ISA [9] rather than reproducing them.

Flan-T5-Base. This group covers supervised fine-tuning methods based on Flan-T5-Base, including THOR [6], RVISA [8], and MT-ISA [9]. We reproduce THOR using its publicly available code (the only method among these with released implementation), enabling additional evaluation on the ISA subsets, while reporting the published results for the other methods. Using the same codebase, we also reproduce the *Vanilla Inst-FT* baseline with the instruction *"Given the sentence {text}, the sentiment polarity towards {aspect_term} is"*.

LLMs with LoRA. For broader comparison, we fine-tune Llama-3.1-8B-Instruct, Qwen3-8B, and Flan-T5-XXL (11B) using instruction-based supervised fine-tuning. The input prompt follows the same template as the *Vanilla Inst-FT* baseline described above. This setting is reported as *Vanilla Inst-FT* in Table 4. To reduce computational cost, we apply Low-Rank Adaptation (LoRA) with $r = 16$, $\alpha = 32$, and a dropout rate of 0.05 across all models, training them for 4 epochs with a learning rate of 2×10^{-4} in `bfloat16`. Specifically, LoRA is applied to all linear layers for Llama and Qwen, and to the attention projection matrices for Flan-T5-XXL.

5.5 Implementation Details

Hyperparameters. Table 3 summarizes the hyperparameter configuration for training. We apply different learning rates, using a smaller rate for the pre-trained Flan-T5 encoder (Backbone LR) and a larger rate for the task-specific projection and classification

[4] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
[5] https://huggingface.co/Qwen/Qwen3-8B

layers (Head LR). Fixed class weights follow the class distributions in the training sets. For $\mathcal{L}_{\text{subj}}$, α_1 and β_1 weight the Neutral and Polar classes, respectively, and are shared across domains due to their similar class ratios. For $\mathcal{L}_{\text{pol}}$, α_2 and β_2 weight the Positive and Negative classes, respectively, and are set separately for each domain to account for differing polarity distributions.

Table 3. Hyperparameter configuration for training MASS with Flan-T5-Base.

Hyperparameter	**Value**
Max Source Length	128
Batch Size	16
Epochs	8
Backbone LR	2×10^{-5}
Head LR	5×10^{-4}
Projection Dimension	256
Projection Dropout	0.2
Subjectivity Class Weights	$[\alpha_1, \beta_1] = [2.5,\ 0.6]$
Restaurant Polarity Weights	$[\alpha_2, \beta_2] = [1.0,\ 2.5]$
Laptop Polarity Weights	$[\alpha_2, \beta_2] = [1.0,\ 1.0]$
Orthogonality Weight	$\gamma = 0.5$
Label Smoothing	0.1
Optimizer	AdamW
Weight Decay	0.01
LR Scheduler	Linear (10% warm-up)
Random Seeds	13, 42, 2024

Flan-T5-XXL Configuration. As full fine-tuning Flan-T5-XXL is computationally prohibitive, we load it in `bfloat16` during training and apply LoRA to the query (`q`) and value (`v`) attention matrices, with $r = 16$, $\alpha = 32$, and dropout 0.05. The subspace projection input dimension is increased to 4096, while all other settings follow the Base configuration. This identical LoRA backbone setup ensures a strictly controlled comparison against the *Flan-T5-XXL + Vanilla Inst-FT* baseline.

6 Result Analysis

6.1 Overalls Results

As shown in Table 4, scaling MASS from Flan-T5-Base to Flan-T5-XXL consistently improves performance across both domains and evaluation settings. The larger variant yields gains of 2.61–10.08 points across the reported metrics, with especially substantial improvements on the ISA subsets, demonstrating that MASS continues to benefit effectively from increased backbone capacity. Among methods built on the Flan-T5-Base variant, MASS outperforms the strongest baseline, MT-ISA, by 1.02 Macro-F1

Table 4. Main results compared with baselines on the Restaurant and Laptop datasets [10]. All results are averaged across 3 random seeds, except for those taken from prior studies.

Method	Restaurant				Laptop			
	All-Acc.	All-MF1	ISA-Acc.↑	ISA-MF1	All-Acc.	All-MF1	ISA-Acc.	ISA-MF1
Training-free Baselines								
GPT-4.1 + CoT	87.86	76.51	68.91	68.11	80.88	74.43	66.86	66.80
GPT-4.1 + Zero-shot	88.39	77.17	70.04	69.13	80.56	73.48	63.43	63.46
GPT-4.1 + Few-shot	88.46	80.13	72.41	72.94	82.45	77.41	69.71	69.57
Supervised Fine-tuning Approaches								
BERT-Base + SPC† [4]	83.57	77.16	65.54	–	78.22	73.45	69.54	–
BERT-Base + ADA† [17]	87.14	80.05	65.92	–	78.96	74.18	70.11	–
BERT-Base + RGAT† [22]	86.60	81.35	67.79	–	78.21	74.07	72.99	–
BERT-Base$_{Asp}$ + CEPT† [10]	87.50	82.07	67.79	–	81.66	78.38	75.86	–
T5-Base + ABSA-ESA† [13]	88.29	81.74	70.78	–	82.44	79.34	80.00	–
BERT-Base$_{Asp}$ + SCAPT† [10]	89.11	83.79	72.28	–	82.76	79.15	77.59	–
Flan-T5-Base + THOR [6]	86.16	77.98	64.42	64.69	80.72	76.70	73.14	69.07
Flan-T5-Base + Vanilla Inst-FT [6]	86.52	79.10	64.79	65.31	81.03	77.39	75.43	71.29
Flan-T5-Base + RVISA$^{\star}_{v}$ [8]	86.43	78.49	65.92	–	80.72	76.49	73.71	–
Flan-T5-Base + MT-ISA$^{\dagger}_{O}$ [9]	85.89	77.85	66.29	–	81.98	78.10	75.43	–
Flan-T5-Base + RVISA$^{\star}_{g}$ [8]	86.61	78.92	66.67	–	81.19	77.13	75.43	–
Flan-T5-Base + MT-ISA$^{\dagger}_{I}$ [9]	88.21	82.45	70.41	–	**82.91**	79.86	80.57	–
Flan-T5-Base + MASS (Ours)	**88.35**	**83.47**	**74.87**	**73.21**	82.19	**80.34**	**81.67**	**75.08**
Llama-3.1-8B-IT + Vanilla Inst-FT	89.11	82.53	76.78	77.07	82.60	78.74	77.14	75.17
Qwen3-8B + Vanilla Inst-FT	90.62	85.03	77.53	77.15	83.39	80.12	80.57	77.60
Flan-T5-XXL + Vanilla Inst-FT	90.89	85.68	80.15	80.01	85.42	82.67	86.29	84.63
Flan-T5-XXL + MASS (Ours)	**91.16**	**86.27**	**82.77**	**81.88**	**85.89**	**82.95**	**87.43**	**85.16**

Scores marked with $\star$ are taken from RVISA [8], while all †-marked results, including the compiled baselines, are taken from MT-ISA [9]. Subscripts v/g denote RVISA rationales generated by Vicuna-13B/GPT-3.5-turbo; I/O denote MT-ISA confidence weighting applied to the input embedding/auxiliary loss.

points on Restaurant All and 0.48 points on Laptop All. The gains are more pronounced on ISA accuracy, reaching 4.46 points on Restaurant and 1.10 points on Laptop, indicating the particular effectiveness of MASS for implicit sentiment instances. We do not directly compare MASS with Flan-T5-XXL against previously reported baselines because those methods use full fine-tuning, whereas our XXL model is fine-tuned with LoRA under computational constraints.

Beyond the baselines implemented with the same backbone, we further compare MASS with approaches built on different models for the same task. For base-sized models, MASS with Flan-T5-Base consistently outperforms methods based on BERT-Base and T5-Base, which have broadly comparable model sizes, with particularly clear gains on the ISA subsets. Similarly, at the larger-model scale, MASS with Flan-T5-XXL achieves stronger results than LoRA prompt-based fine-tuned Llama-3.1-8B-Instruct and Qwen3-8B, especially for ISA. Specifically, it improves Macro-F1 by 4.81 and 4.73 points over the two models on Restaurant ISA, and by 9.99 and 7.56 points on Laptop ISA, respectively. Moreover, when compared to the vanilla instruction-tuned Flan-T5-XXL baseline, Flan-T5-XXL with MASS consistently achieves superior performance across all metrics, yielding gains of 1.87 and 0.53 Macro-F1 points on Restaurant and

Laptop ISA subsets, respectively. Finally, both Flan-T5-Base and Flan-T5-XXL variants of MASS outperform the GPT-based prompting baselines, whose performance drops substantially on the ISA subsets—particularly Laptop ISA—highlighting the limitations of prompting alone for implicit sentiment reasoning.

6.2 Statistical Significance Testing

Because MT-ISA [9], the strongest existing baseline, does not publicly release its source code, we cannot conduct significance testing against it. We therefore select THOR [6], the most recent baseline with publicly available code, for statistical comparison.

We assess whether the accuracy differences among MASS, Vanilla Inst-FT, and THOR with Flan-T5-Base are statistically significant using paired predictions on the same test instances. We first apply Cochran's Q test [3] to test the null hypothesis that the three methods have equal accuracy ($df = 2$, $\alpha = 0.05$). For splits where this hypothesis is rejected, we conduct pairwise McNemar tests [12]. Since three pairwise comparisons are performed—MASS vs. THOR, MASS vs. Vanilla Inst-FT, and THOR vs. Vanilla Inst-FT—we use a Bonferroni-adjusted threshold of $\alpha_{\mathrm{Bonf}} = 0.05/3 = 0.0167$ [2] to control the family-wise error rate. All pairwise McNemar tests use the chi-squared approximation with Yates's continuity correction.

Table 5. Cochran's Q test for accuracy differences among MASS, Vanilla Inst-FT, and THOR.

Split	Q	p**-value**	**Decision**
Restaurant ISA	17.6047	0.00015	Reject H_0
Restaurant All	7.6863	0.02143	Reject H_0
Laptop ISA	6.0694	0.04813	Reject H_0
Laptop All	1.5701	0.45610	Fail to reject H_0

H_0: all three methods have equal classification accuracy. Results are evaluated at $\alpha = 0.05$ with $df = 2$.

Tables 5 and 6 report the omnibus and post-hoc results, respectively. Cochran's Q test identifies significant accuracy differences on the Restaurant All, Restaurant ISA, and Laptop ISA splits, whereas no significant difference is observed on Laptop All. Accordingly, post-hoc McNemar tests are conducted only on the three significant splits. The results confirm that MASS significantly outperforms both THOR and Vanilla Inst-FT across all these splits. Nevertheless, THOR is not among the top-performing baselines in our experiments; therefore, these significance results should be interpreted as supporting evidence rather than conclusive proof that MASS is the best-performing method.

6.3 Ablation Study

We evaluate the contribution of each MASS component using Flan-T5-Base. As shown in Table 7, removing aspect markers consistently degrades performance, with the largest

Table 6. Post-hoc pairwise comparisons using McNemar's test with Bonferroni correction.

Split	Pairwise Result	Discordant	p-value
Restaurant All	MASS > THOR*	70	0.00281
	MASS > V-FT*	73	0.00497
	THOR vs V-FT (ns)	37	1.00000
Restaurant ISA	MASS > THOR*	58	0.00009
	MASS > V-FT*	61	0.00021
	THOR vs V-FT (ns)	28	1.00000
Laptop ISA	MASS > THOR*	46	0.01245
	MASS > V-FT*	49	0.01532
	THOR vs V-FT (ns)	31	0.85610

*: significant after Bonferroni correction ($p < \alpha_{\mathrm{Bonf}} = 0.0167$). Pairwise tests are reported only for splits with a significant Cochran's Q result. V-FT denotes Vanilla Inst-FT.

drop in Laptop ISA Accuracy (81.67% to 74.29%), suggesting that explicit marking supports more relevant contextual representations. Replacing the two subspaces with a single three-class classifier causes the largest overall degradation, reducing Restaurant ISA Macro-F1 by 9.82 points; thus, the two-subspace decomposition shows the largest observed contribution. Removing class weights also substantially reduces performance; for example, Laptop ISA Macro-F1 decreases from 75.08% to 68.51%, highlighting the benefit of accounting for label imbalance. In contrast, removing the orthogonality penalty lowers performance on six of the eight metrics but slightly improves Restaurant ISA Accuracy and Laptop ISA Macro-F1, suggesting a smaller and less consistent contribution. Finally, replacing the start-marker representation with either the end-marker representation or mean span pooling also reduces ISA Macro-F1 in both domains.

7 Revisiting Prior Limitations with MASS

We next examine whether MASS alleviates the limitations identified for THOR in Section 2. On the full test sets, MASS increases AAR to 6.71% on Laptop (+1.72 pp over THOR) and 6.76% on Restaurant (+0.22 pp), indicating a positive but domain-dependent improvement in aspect focus. The confusion matrices in Table 8 show that MASS reduces the total number of errors in both domains, for example from 155 to 133 on Restaurant. However, Neutral-involved errors account for a larger proportion of the remaining mistakes under MASS (89.47–93.58%; 91.32% pooled) than under THOR (83.8% pooled). This increase in proportion does not indicate more Neutral-related errors in absolute terms; rather, MASS reduces Positive–Negative confusions more substantially than errors involving Neutral. This pattern is consistent with the All-MF1 improvements reported in Table 4 (+5.49 pp on Restaurant and +6.25 pp on Laptop), suggesting that the overall gains arise mainly from fewer polarity errors, while the Neutral–Polar boundary remains challenging. **Conclusion:** MASS improves overall sentiment classification and aspect focus, although the latter varies by domain. It miti-

Table 7. Component ablations and marker-representation variants of MASS on the Restaurant and Laptop datasets using Flan-T5-Base.

Variant	Restaurant				Laptop			
	All		ISA		All		ISA	
	Acc.	MF1	Acc.	MF1	Acc.	MF1	Acc.	MF1
Full Method	**88.35**	**83.47**	74.87	**73.21**	82.19	**80.34**	**81.67**	75.08
w/o Aspect Marker	87.50	81.23	72.66	71.13	80.56	76.95	74.29	71.03
w/o Orthogonality	88.13	82.75	**75.28**	72.66	81.50	78.96	81.14	**75.42**
w/o Two Subspaces	86.07	77.35	63.67	63.39	81.50	77.36	73.71	70.46
w/o Class Weights	86.88	78.65	67.42	67.66	80.25	75.15	70.86	68.51
End-Marker Rep.	86.79	80.54	73.53	70.58	**82.45**	79.37	79.24	73.50
Mean Span Pooling	87.59	81.79	74.33	72.63	82.14	79.32	79.19	73.89

Full Method uses the hidden state at `<extra_id_0>`. Without the aspect marker (w/o Aspect Marker), the representation is obtained by mean pooling over all non-padding prompt tokens. End-Marker Rep. uses `<extra_id_1>`, while Mean Span Pooling averages the hidden states from `<extra_id_0>` to `<extra_id_1>`, inclusive.

Table 8. Confusion matrices for MASS with Flan-T5-Base on the full test sets. Rows are true labels (T), columns are predicted labels (P).

MASS – Restaurant

T/P	Neg	Neu	Pos
Neg	**167**	24	5
Neu	22	**121**	53
Pos	9	20	**699**

MASS – Laptop

T/P	Neg	Neu	Pos
Neg	**105**	23	0
Neu	22	**132**	15
Pos	7	42	**292**

gates but does not eliminate the Neutral–Polar ambiguity that dominates the remaining errors.

8 Conclusion

In this paper, we proposed MASS, a novel framework that enhances ISA by explicitly marking aspect terms and decomposing representations into distinct subjectivity and polarity subspaces. Without relying on costly LLM-based data augmentation, MASS achieves superior performance on English Laptop and Restaurant benchmarks, particularly on challenging ISA subsets. While effective, our framework presents several limitations, including domain and language constraints, architectural dependence on Flan-T5, potential error propagation in hierarchical inference, and limited calibration generality under compute-constrained setups. To address these challenges, future work will focus on expanding MASS to cross-domain and multilingual settings, refining hierar-

chical routing to mitigate error cascade, exploring broader encoder architectures, and developing more robust confidence calibration methods.

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